

EE3001 Foundations of Data Engineering

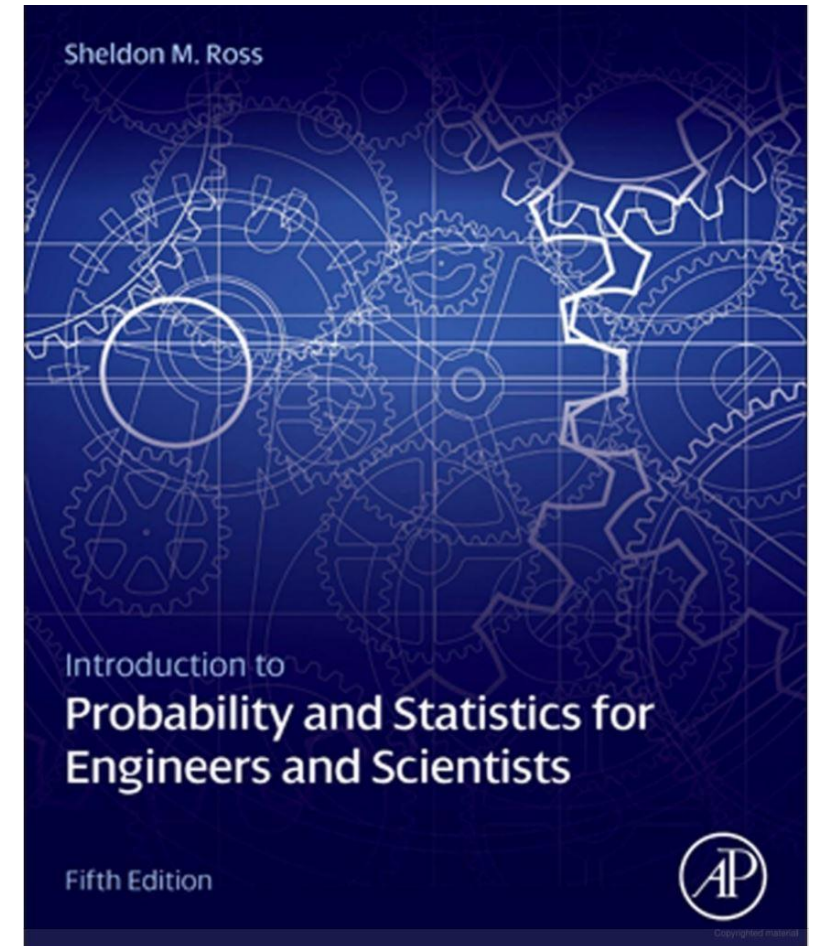
Prof. YUEN Shiu Yin, Kelvin

Department of Electrical Engineering, City University of Hong Kong

Web page: <https://www.ee.cityu.edu.hk/~syyuen>

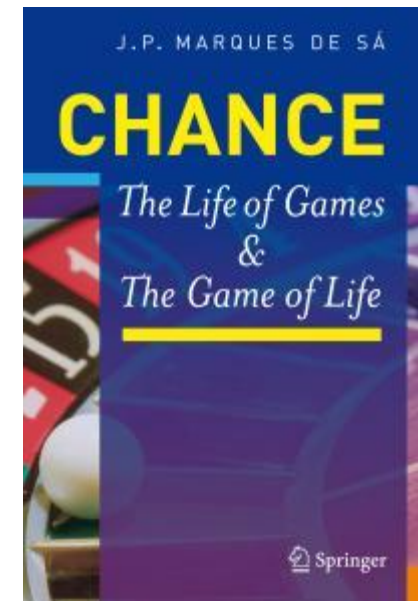
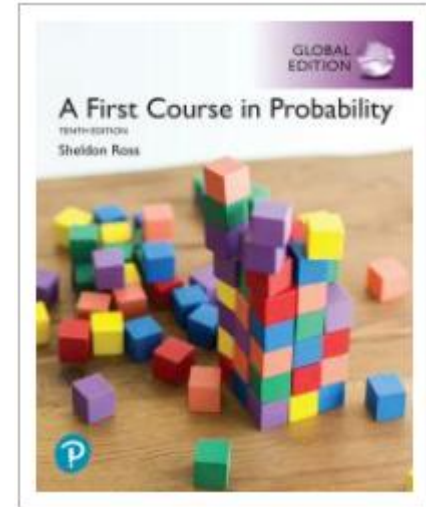
Text book

- Introduction to Probability and Statistics for Engineers and Scientists, S.M. Ross 5nd Ed. (2014)
- The online version of the book is available in CityU library.
- Download the full PDF of the book NOW



References

- A First Course in Probability, S. Ross, 10th Ed. (2020)
- Chance, The Life of Games & The Game of Life, J.P. Marques De Sá (2008)
- The online version of the two books is available in CityU library



PPT

- The ppt may be challenging to read for some of you if your mathematical background is weak
- Therefore, important to attend class and listen to the explanations and ask questions
- The ppt is prepared using the content of the text book. **Form the good habit of reading the text book** if you do not fully understand the lecture

Practice questions and practice questions with answers

- Practice questions are used for self-testing at home to clarify understanding
- Practice questions with answers are used as illustrative examples in class as well as for self understanding of concepts
- You can use it in both ways, for example, understand the solution process and then see whether you can do the same question without the answer

Topics

Probability

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Statistics

- Topic 1: Probability
- Topic 2: Bayes' rule
- Topic 3: Detecting independent events
- Topic 4: Random variables and expectation
- Topic 5: Special random variables

- Topic 6: Descriptive Statistics
- Topic 7: Parameter Estimation
- Topic 8: Hypothesis testing
- Topic 9: Non-parametric statistics
- Topic 10: Regression and prediction

Assessment

- Test 1 (18%) Test 2 (18%) In-class Assignments (10%) Team-based Learning Assignment (4%) Exam (50%)
- To pass the course, students are required to achieve at least 30 out of 100 in coursework and 30 out of 100 in the examination
- Please mark your diary now
- Test 1 Wk 6 Scope: everything taught in wk 1 - 5
- Test 2 Wk 11 Scope: everything taught in wk 6 – 10
- Late hand in of assignments will get no mark
- Non-attendance in test will get no mark (department guideline)
- Familiarize with how to use the statistical functions in your calculator
- An appendix consisting of a set of formulae and tables will be given in tests and exam

Use of AI in Teaching

- There will be periods in the lecture where you are asked to help AI to help you understand/prepare for in class assignments
- Accessing Copilot in CityU

<https://www.cityu.edu.hk/its/news-events/news/2024/09/16/microsoft-copilot-enterprise-now-available-all-staff-and-students>

Note: You are protected by “enterprise data protection”

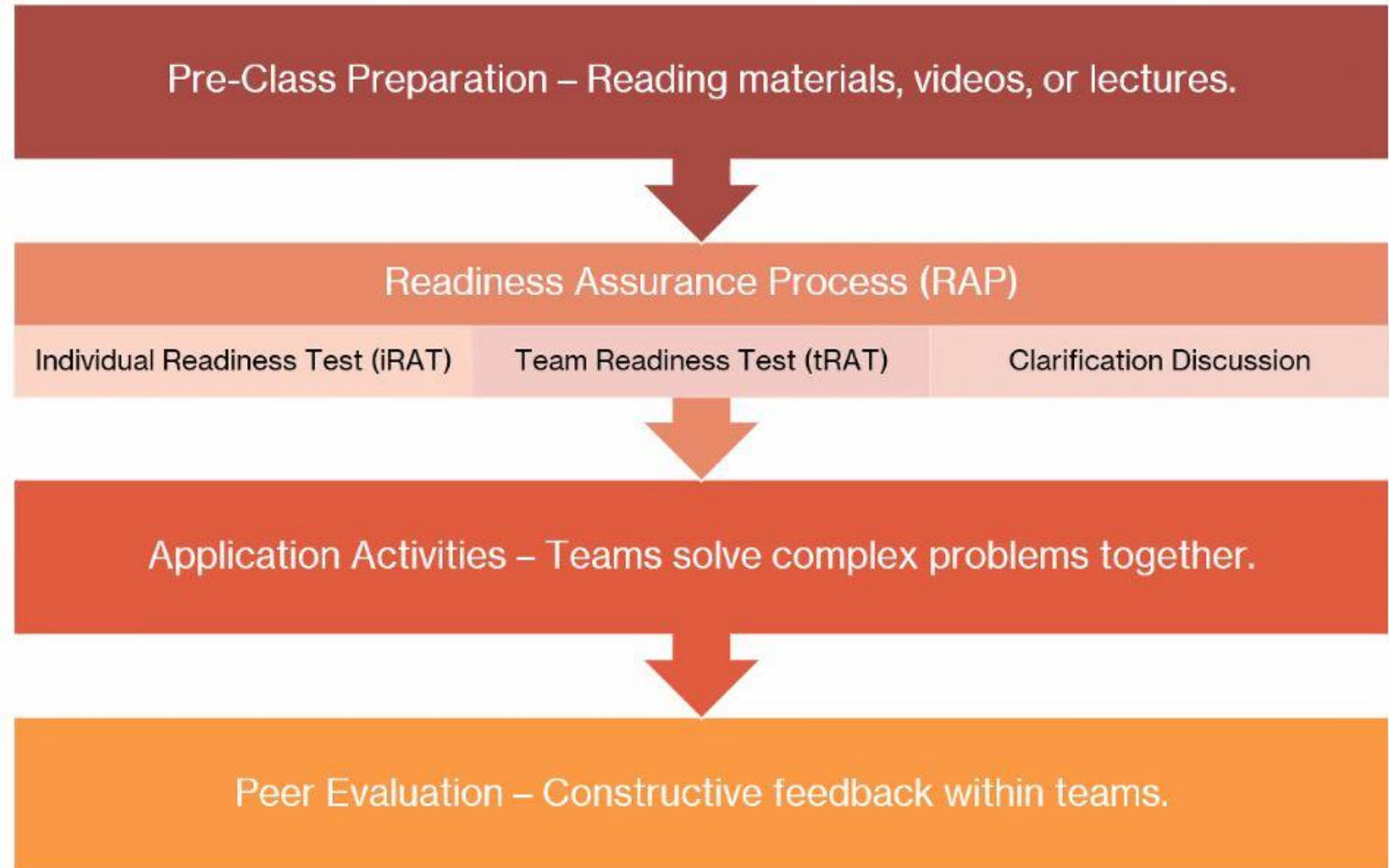
In class Assignment Format

- In class, **so you must attend in order to get marks**
- Closed book. Write your answer on paper
- Hand in scanned answer with name and SID through Canvas at the end of lecture
- Only hand in your answer. Don't hand in on behalf of others (impersonation is an offence)
- TA will mark it within 1 week
- I'll discuss answers at the beginning of next lecture
- Any mark correction request must be made by Friday. No late request would be entertained

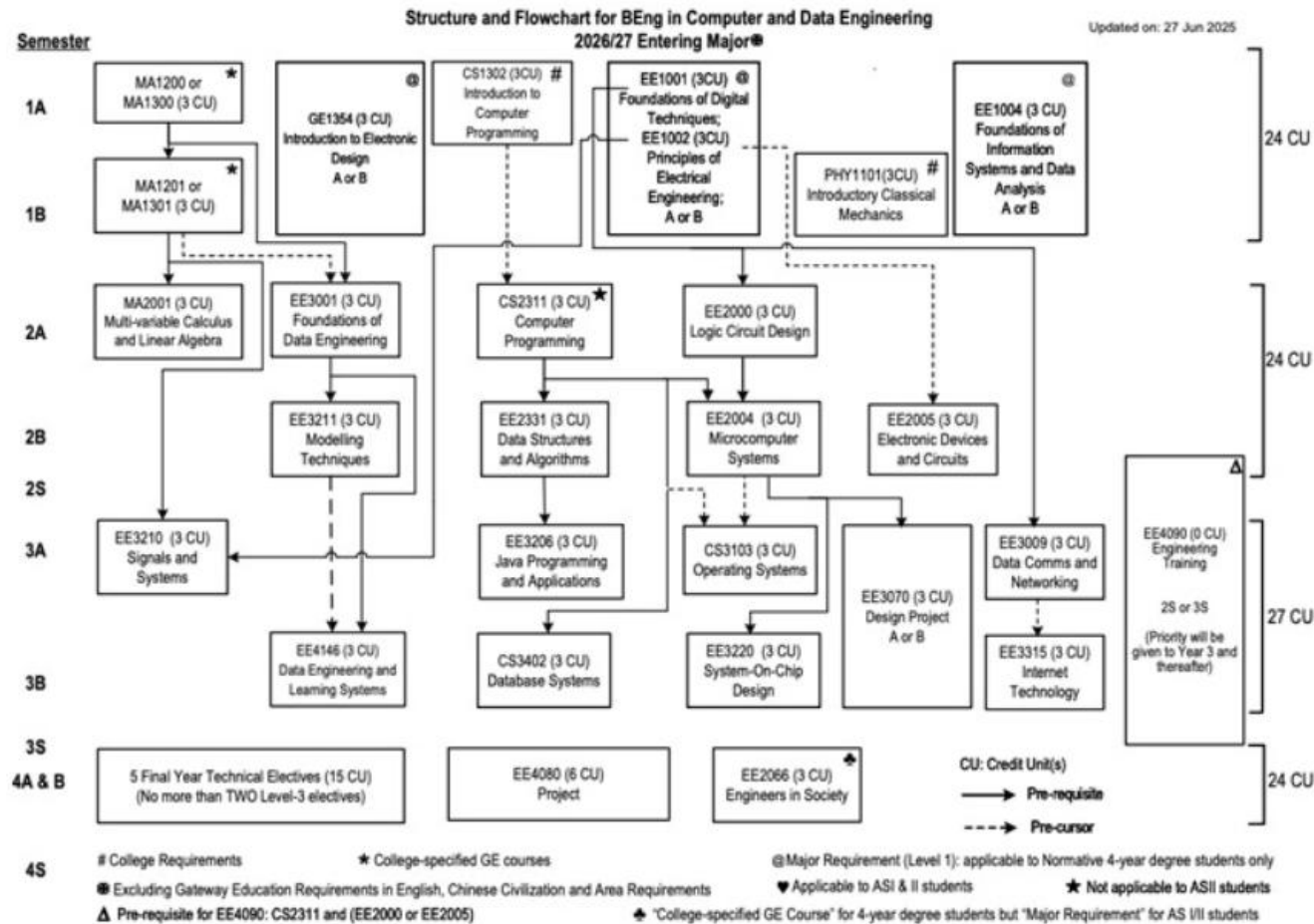
Team-based learning (TBL) assignment

- There will be one team-based learning assignment at Wk 13
- You can freely form a team
- You will be asked to solve a real problem using the knowledge that you have learnt
- Individual and team readiness test (iRAT, tRAT) and clarification discussion will be done in Wk 11 and 12
- Peer evaluation will be done immediately after your group hand in your TBL assignment at Wk 13
- Freely from groups of 3. You may like to form a study group of 3 or more now

The TBL Process



Relationships with other subjects



- EE1004 bring up to M1 level
- EE3211 about statistics application. EE3001 provides mathematical background
- EE4146 about data mining and machine learning. EE3001 provides mathematical background
- Relations to all other subjects: probability and statistics are fundamental

Weekly TA Help

- If you have questions, ask the TA responsible for the week through email
- They would answer you promptly

TA Name	TA Email	Responsible weeks
ZHANG Keyang	keyazhang4-c@my.cityu.edu.hk	Wk 1 - 6 Wk 13 and revision weeks
QU Fuchuan	fuchuanqu2-c@my.cityu.edu.hk	Wk 7 - 12 Wk 13 and revision weeks